CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 13: Statistics

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Mean of grouped data

1. The data below has a mean of 35 and a total frequency of 25. Find x and y. CBSE 2026, 5 marks

Class	0–10	10–20	20–30	30–40	40–50	50–60	60–70
Frequency	1	x	5	7	y	3	1

SOLUTION

Total frequency: $17 + x + y = 25$, so $x + y = 8$ ... (1)

Use the class marks 5, 15, 25, 35, 45, 55, 65.

$\sum f_i x_i = 5 + 15x + 125 + 245 + 45y + 165 + 65 = 605 + 15x + 45y$

Mean $= \dfrac{605 + 15x + 45y}{25} = 35$, so $15x + 45y = 270$, that is $x + 3y = 18$ ... (2)

Subtracting (1) from (2): $2y = 10$, so $y = 5$ and $x = 3$.

Answer: $x = 3$, $y = 5$

2. The mean of the data below is 18. Find the missing frequency f, and then find the mode. CBSE 2025, 5 marks

Class	11–13	13–15	15–17	17–19	19–21	21–23	23–25
Frequency	7	6	9	13	f	5	4

SOLUTION

Use the class marks 12, 14, 16, 18, 20, 22, 24.

$\sum f_i = 44 + f$

$\sum f_i x_i = 84 + 84 + 144 + 234 + 20f + 110 + 96 = 752 + 20f$

$\dfrac{752 + 20f}{44 + f} = 18$, so $752 + 20f = 792 + 18f$ and $f = 20$.

The highest frequency is now 20, so the modal class is 19–21 with $l = 19$, $h = 2$, $f_1 = 20$, $f_0 = 13$, $f_2 = 5$.

Mode $= 19 + \dfrac{20 - 13}{40 - 13 - 5} \times 2 = 19 + \dfrac{14}{22}$

Answer: $f = 20$; **mode** ≈ 19.64

3. Observations $x_1, x_2, \ldots, x_n$ have frequencies $f_1, f_2, \ldots, f_n$ and mean $\bar{x}$. What is the value of $\sum f_i(x_i - \bar{x})$? (a) $n\bar{x}$ (b) 1 (c) $\sum f_i$ (d) 0 CBSE 2024, 1 mark

SOLUTION

$\sum f_i(x_i - \bar{x}) = \sum f_i x_i - \bar{x} \sum f_i$

Since $\bar{x} = \dfrac{\sum f_i x_i}{\sum f_i}$, we have $\bar{x} \sum f_i = \sum f_i x_i$, so the value is 0.

Answer: (d)

4. Every observation in a set of data is increased by 3. What happens to the mean? (a) it does not change (b) it increases by 3 (c) it increases by 6 (d) it increases by $3n$ CBSE 2023, 1 mark

SOLUTION

The total increases by $3n$, so the mean increases by $\dfrac{3n}{n} = 3$.

Answer: (b)

5. The table shows what 30 employees of a company spend on lunch each day. Find the mean daily spending. CBSE 2022 Term 2, 3 marks

Daily spending (₹)	100–120	120–140	140–160	160–180	180–200
Number of employees	8	3	8	6	5

SOLUTION

Class marks: 110, 130, 150, 170, 190.

$\sum f_i x_i = 880 + 390 + 1200 + 1020 + 950 = 4440$ and $\sum f_i = 30$

Mean $= \dfrac{4440}{30}$

Answer: ₹148

6. The mean of this data is 18. Find the frequency f of the class 19–21. CBSE 2018, 4 marks

Class	11–13	13–15	15–17	17–19	19–21	21–23	23–25
Frequency	3	6	9	13	f	5	4

SOLUTION

Use the class marks 12, 14, 16, 18, 20, 22, 24.

$\sum f_i = 40 + f$ and $\sum f_i x_i = 36 + 84 + 144 + 234 + 20f + 110 + 96 = 704 + 20f$

$\dfrac{704 + 20f}{40 + f} = 18$, so $704 + 20f = 720 + 18f$

Answer: $f = 8$

2 Empirical relationship

7. The mean of some data is 12 and its mode is 21. Find the median. (a) 6 (b) 13.5 (c) 15 (d) 14 CBSE 2026, 1 mark

SOLUTION

$3 \times \text{Median} = \text{Mode} + 2 \times \text{Mean} = 21 + 24 = 45$

Median $= 15$

Answer: (c)

8. For a set of data, the mode is $15x$ and the mean is $18x$. Find the median. CBSE 2025, 1 mark

SOLUTION

$3 \times \text{Median} = \text{Mode} + 2 \times \text{Mean} = 15x + 36x = 51x$

Answer: Median $= 17x$

3 Median of grouped data

9. When data is arranged in order, the observation in the middle is called the: (a) mode (b) median (c) mean (d) deviation CBSE 2024, 1 mark

SOLUTION

The median is the middle value of data arranged in order.

Answer: (b)

10. In a game of BINGO the host has balls numbered 1 to 75 and calls out the number on a ball picked at random. In one game 48 balls were called, as shown in the table. (i) Name the median class. (ii) Find the probability that the first ball called had an even number. (iii) Find the median of the data, OR find its mode. CBSE 2024, 4 marks, case study

Numbers announced	0–15	15–30	30–45	45–60	60–75
Number of times	8	9	10	12	9

SOLUTION

(i) $n = 48$, so $\frac{n}{2} = 24$. The cumulative frequencies are 8, 17, 27, 39, 48. The first one greater than 24 is 27, so the median class is 30–45.

(ii) Of the numbers 1 to 75, the even numbers are $2, 4, \ldots, 74$, which is 37 numbers. Probability $= \frac{37}{75}$

(iii) Median $= l + \frac{\frac{n}{2} - cf}{f} \times h = 30 + \frac{24 - 17}{10} \times 15 = 30 + 10.5 = 40.5$

OR: the modal class is 45–60, with $f_1 = 12$, $f_0 = 10$, $f_2 = 9$. Mode $= 45 + \frac{12 - 10}{24 - 10 - 9} \times 15 = 45 + 6 = 51$

Answer: (i) 30–45 (ii) $\frac{37}{75}$ **(iii) median 40.5, or mode 51**

11. The table shows the monsoon rainfall recorded in 24 sub-divisions of the country. (i) Write the modal class. (ii) Find the median, OR find the mean rainfall. (iii) A sub-division with at least 1000 mm of rain is said to have good rainfall. How many had good rainfall? CBSE 2023, 4 marks

Rainfall (mm)	200–400	400–600	600–800	800–1000	1000–1200	1200–1400	1400–1600	1600–1800
Sub-divisions	2	4	7	4	2	3	1	1

SOLUTION

(i) The highest frequency is 7. **Modal class: 600–800.**

(ii) Cumulative frequencies: 2, 6, 13, 17, 19, 22, 23, 24. $n = 24$, so $\frac{n}{2} = 12$ and the median class is 600–800, with $l = 600$, $cf = 6$, $f = 7$, $h = 200$.

Median $= 600 + \frac{12 - 6}{7} \times 200 = 600 + 171.43$. **Median** ≈ 771.43 **mm.**

OR: with class marks 300, 500, ..., 1700, $\sum f_i x_i = 600 + 2000 + 4900 + 3600 + 2200 + 3900 + 1500 + 1700 = 20400$. Mean $= \frac{20400}{24}$. **Mean** $= 850$ **mm.**

(iii) Classes from 1000 mm upwards: $2 + 3 + 1 + 1$. **7 sub-divisions had good rainfall.**

12. The median of this data is 34.5. Find x. CBSE 2022 Term 2, 3 marks

Class	0–10	10–20	20–30	30–40	40–50	50–60	60–70
Frequency	3	5	11	10	x	3	2

SOLUTION

$n = 34 + x$. The median 34.5 lies in the class 30–40, so $l = 30$, $f = 10$, $h = 10$ and $cf = 3 + 5 + 11 = 19$.

$34.5 = 30 + \frac{\frac{34+x}{2} - 19}{10} \times 10$, so $\frac{34 + x}{2} - 19 = 4.5$

$34 + x = 47$

Answer: $x = 13$

13. The median of this data is 32.5 and the total frequency is 40. Find f_1 and f_2. CBSE 2019, 4 marks

Class	0–10	10–20	20–30	30–40	40–50	50–60	60–70
Frequency	f_1	5	9	12	f_2	3	2

SOLUTION

Total: $f_1 + f_2 + 31 = 40$, so $f_1 + f_2 = 9$ ... (1)

The median 32.5 lies in 30–40, so $l = 30$, $f = 12$, $h = 10$, $cf = f_1 + 14$ and $\frac{n}{2} = 20$.

$32.5 = 30 + \frac{20 - (f_1 + 14)}{12} \times 10$, so $2.5 \times 12 = 10(6 - f_1)$ and $6 - f_1 = 3$

$f_1 = 3$, and from (1) $f_2 = 6$.

Answer: $f_1 = 3$, $f_2 = 6$

14. The table shows the salaries of 280 people. Find the median salary. CBSE 2018, 3 marks

Salary (thousand ₹)	5–10	10–15	15–20	20–25	25–30	30–35	35–40	40–45	45–50
Number of people	49	133	63	15	6	7	4	2	1

SOLUTION

$n = 280$, so $\frac{n}{2} = 140$. The cumulative frequencies are 49, 182, ..., so the median class is 10–15.

$l = 10$, $cf = 49$, $f = 133$, $h = 5$

Median $= 10 + \frac{140 - 49}{133} \times 5 = 10 + \frac{455}{133} \approx 10 + 3.42$

Answer: about 13.42 thousand rupees, that is about ₹13,420

4 Mode of grouped data

15. The table gives the marks of 80 students in a test. Find the modal class. (a) 10–20 (b) 20–30 (c) 30–40 (d) 50–60 CBSE 2023, 1 mark

Marks	Less than 10	Less than 20	Less than 30	Less than 40	Less than 50	Less than 60
Number of students	3	12	27	57	75	80

SOLUTION

The table is cumulative. Subtract to get the frequency of each class:

Class	0–10	10–20	20–30	30–40	40–50	50–60
Frequency	3	9	15	30	18	5

The highest frequency is 30, in the class 30–40.

Answer: (c)

16. Find the mode of this data. CBSE 2022 Term 2, 2 marks

Class	10–20	20–30	30–40	40–50	50–60
Frequency	15	10	12	17	4

SOLUTION

The highest frequency is 17, so the modal class is 40–50: $l = 40$, $h = 10$, $f_1 = 17$, $f_0 = 12$, $f_2 = 4$.

$$\text{Mode} = 40 + \frac{17 - 12}{34 - 12 - 4} \times 10 = 40 + \frac{50}{18}$$

Answer: Mode ≈ 42.78

17. Find the mode of this data. CBSE 2019, 3 marks

Class	0–10	10–20	20–30	30–40	40–50	50–60	60–70
Frequency	8	10	10	16	12	6	7

SOLUTION

The highest frequency is 16, so the modal class is 30–40: $l = 30$, $h = 10$, $f_1 = 16$, $f_0 = 10$, $f_2 = 12$.

$$\text{Mode} = 30 + \frac{16 - 10}{32 - 10 - 12} \times 10 = 30 + 6$$

Answer: Mode $= 36$

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